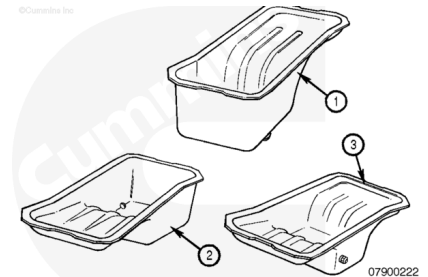


Trainings ▾

General Information

A front sump (1), rear sump (2), or center sump (3) lubricating oil pan can be used, depending on the application. The mounting of the lubricating oil pick-up tube will vary with the lubricating oil pan used.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

- Disconnect the batteries. Reference the equipment manufacturer service information.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>)
- If equipped, disconnect/remove the lubricating oil heater. Refer to Procedure 007-001 (</qs3/pubsys2/xml/en/procedures/40/40-007-001.html>) in Section 7.

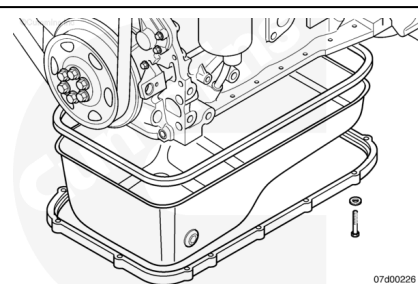
Remove

Rear Gear Train

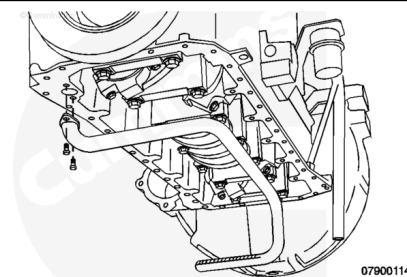
Note : Rear gear train engines use a suspended oil pan that consists of the oil pan, mounting flange, and a flexible gasket.

Remove the oil pan mounting capscrews, oil pan, mounting flange, and flexible gasket.

The lubricating oil pan flange can be damaged if a prybar or screwdriver is used to break the formed-in-place gasket seal. Use a utility knife or the edge of a putty knife to cut the seal and loosen the pan.

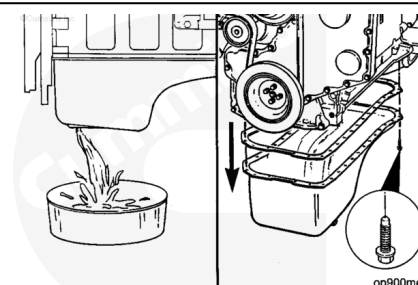


Remove the suction tube, if necessary. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)



Front Gear Train

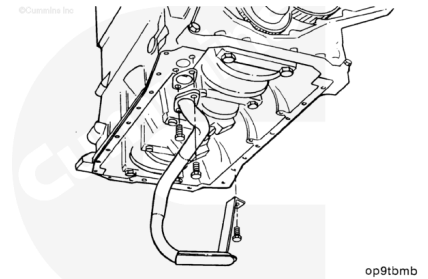
Remove the lubricating oil pan and gasket.



Remove the lubricating oil suction tube, if necessary. Refer to Procedure 007-035 in Section 7.

(/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)

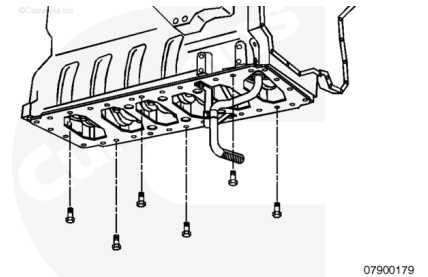
Note : For front gear train engines, the suction tube **must** be removed in order to remove the bedplate.



Remove the bedplate, if equipped.

Remove the capscrews securing the bedplate.

A knife may be required to cut the sealant between the bedplate and the block.



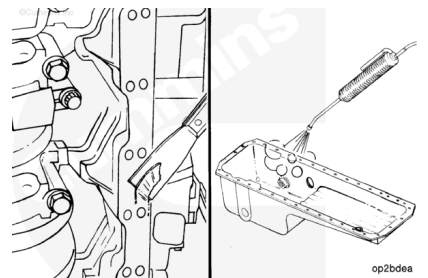
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the gasket surfaces.

Remove all gasket material from the cylinder block and oil pan sealing surface.

Steam clean the oil pan. Dry the oil pan with compressed air.

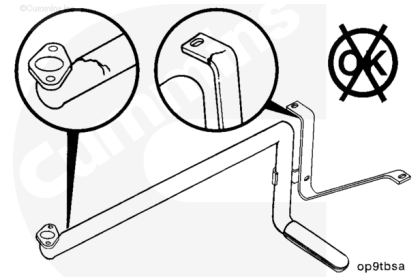
For front gear train engines, degrease the bedplate, if equipped, on both sides.

Inspect the oil pan, suction tube, bedplate, and tube braces for damage.

For suspended oil pans, inspect the mounting flange for damage.

Note : If cracks are found, replace the damaged part.
Do not attempt to repair the oil pan by welding.

Replace the suction tube, if necessary. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)



Clean and check the lubricating oil drain plug threads and sealing surface. Use new sealing washer, if damaged.

Install the lubricating oil pan drain plug.

Torque Value:

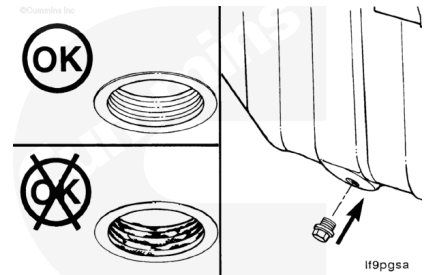
Steel Oil Pan Plug Torque M18
1. 60 n•m [44 ft-lb]

Torque Value:

Steel Oil Pan Plug Torque M22
1. 80 n•m [59 ft-lb]

Torque Value:

Cast Aluminum Oil Pan Plug Torque M22
1. 60 n•m [44 ft-lb]

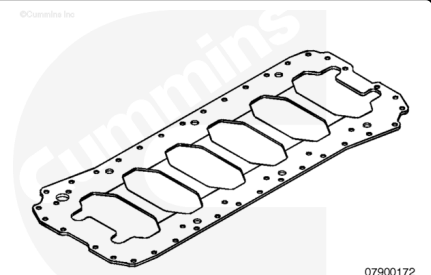


Install

Front Gear Train

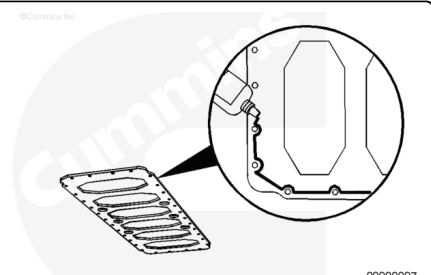
If the engine is equipped with a bedplate, install the bedplate first.

Note : The bedplate will **only** install one way. The center ribs are stamped offset to clear the main bearing caps.

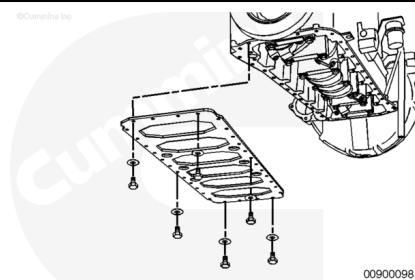


Apply a 1 to 2 mm [0.4 to 0.08 in] bead of RTV, Part Number 3164067, to the block side of the plate.

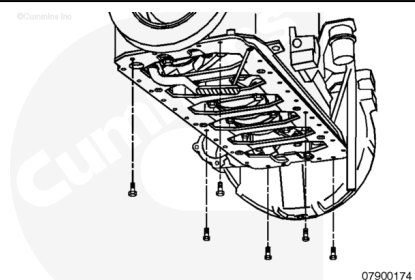
Note : The bedplate **must** be installed within 5 minutes of applying RTV sealant to obtain the proper seal.



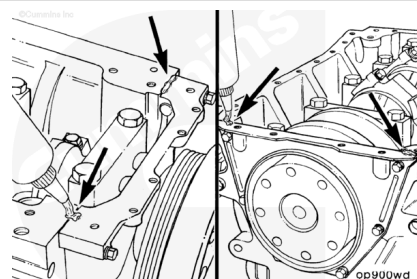
Install the bedplate to the block. Use six oil pan capscrews to hold the bedplate temporarily in place.



Remove the temporary capscrews.

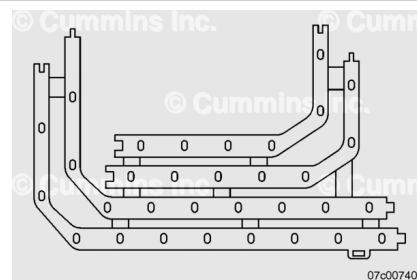


Apply a 2 mm [1/16 in] bead of sealant, Part Number 3164067, to the oil pan mounting surfaces at the cylinder block-to-front gear cover joints, and the cylinder block-to-rear gear housing joints.

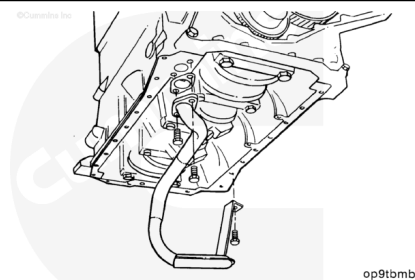


The replacement lubricating oil pan gasket can be in be a four piece segmented lubricating oil pan gasket rather than a single piece lubricating oil pan gasket.

The replacement lubricating oil pan gasket will need to be separated into four pieces and assembled on the lubricating oil pan flange to make a complete lubricating oil pan gasket.



If the suction tube has been removed in order to remove the bedplate, install the suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html>)



Apply a 2-mm [0.063-in] bead of sealant, Part Number 3164067, to the lubricating oil pan flange.

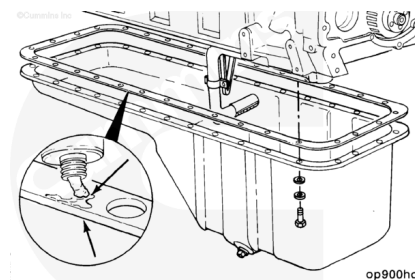
Lay the lubricating oil pan gasket(s) onto the lubricating oil pan flange.

For segmented lubricating oil pan gaskets, ensure the lubricating oil pan gasket segment joints are properly aligned.

Apply a 2-mm [0.063-in] bead of sealant, Part Number 3164067, to the top of the new lubricating pan gasket.

For segmented lubricating oil pan gaskets, make sure the sealant fills the lubricating oil pan gasket segment joints.

Install the lubricating oil pan and lubricating oil pan mounting capscrews.

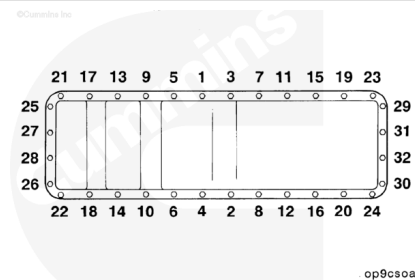


Six-Cylinder

Assemble the washers and capscrews to secure the lubricating oil pan as illustrated.

Tighten all capscrews in the sequence shown in the accompanying chart.

Torque Value: 24 n•m [212 in-lb]

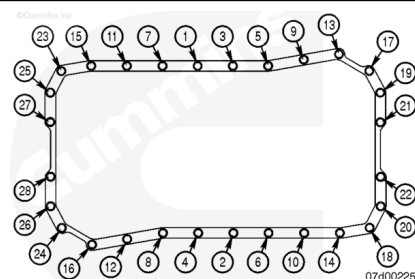


Four-Cylinder

Assemble the washers and capscrews to secure the lubricating oil pan as illustrated.

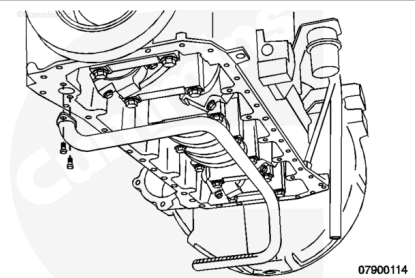
Tighten all capscrews in the sequence shown in the accompanying chart.

Torque Value: 24 n•m [212 in-lb]

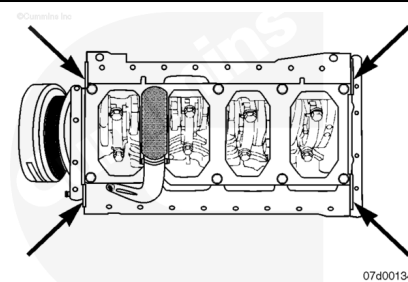


Rear Gear Train

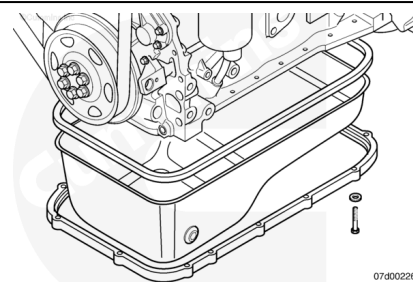
Install the suction tube if has been removed. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-035-tr.html)



Prior to installing the oil pan, apply RTV sealant, Part Number 3164070, to the joints between the lubricating oil pan rail-rear gear housing, and the lubricating oil pan rail-front gear cover.



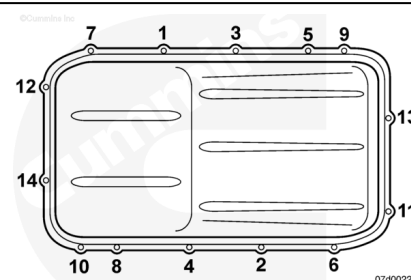
Assemble the new flexible gasket onto the oil pan.
Install the mounting flange over the oil pan/gasket assembly.
Install the oil pan assembly and mounting capscrews.



Tighten all capscrews in the sequence shown.

Torque Value:

1.24 n•m [212 in-lb]



Finishing Steps

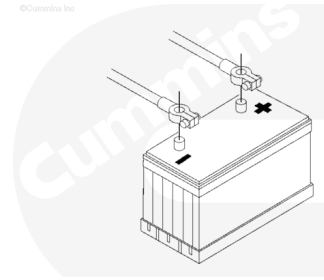
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/40/40-007-037.html)
- Connect the batteries. Reference the equipment manufacturer service information.

- Operate the engine and check for leaks.



13900050

Last Modified: 26-Jan-2016
